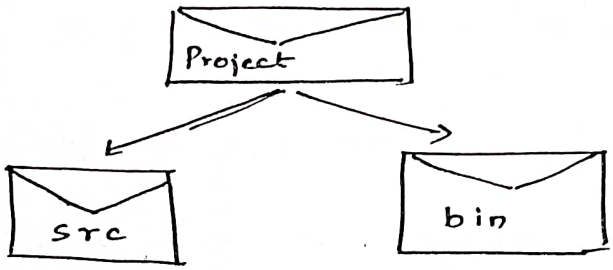
SEGREGATION OF FILES

19/02/2022



STEP 1: Inside project folder create spperate folder for source file and class file. with the name src a bin

STEP 2 : The working directory must be src. ~~file~~ folder

STEP 3 : We can compile the source file with the command:

```
javac -d ../bin fileName.java
```

STEP 4 : We can execute the class file with the command..

```
java -cp ../bin ClassName
```

-d ⇒ indicates direction

.. ⇒ Go back to previous folder / ⇒ route

../bin ⇒ relative path

-cp ⇒ class path.

PACKAGE :

Package is a mechanism which is used to organise & group similar java files together.

Package is a collection of java files. We can create a package with the help of keyword 'package'.

SYNTAX :

```
package PackageName;
```

BUILT-IN PACKAGE :

These are packages which are already created & available for the use are known as Built-in packages.

Eg : java.lang

java.util

java.io

etc,

▶ TO USE THE MEMBER OF ONE PACKAGE IN ANOTHER PACKAGE :

In java we can use public member outside

package.

- Non-public member cannot be used outside the package

▶ **public:**

- It is a keyword
- It is Access Modifier
- Any member which is prefixed with public modif is known as public member
- They can be used outside the package

▶ **public class:**

Any class prefixed with public modifier is known as public class.

To use a class outside the package it is mandatory to make the class as public

The name of the java file should be same as that of public class.

We can use the public member outside the package in two ways:

1) Fully Qualified Name

2) Using 'import' keyword

FULLY QUALIFIED NAME:

The member along with the package name is known as Fully Qualified Name

Eg:

PackageName.ClassName.MemberName;

USING THE 'import' KEYWORD:

'import' is used to load the class from specified package to current package.

'import' is a keyword.

SYNTAX TO IMPORT SPECIFIC CLASS:

```
import PackageName.ClassName;
```

SYNTAX TO IMPORT ALL THE CLASSES OF PACKAGE:

```
import PackageName.*;
```

RULES TO BE FOLLOWED:

1) 'import' statement must be written above the class

2) We can import only public class

3) Non-public class cannot be imported

MODIFIERS:

Modifiers are the keywords which are used to modify behaviours of members of java.

We have two types of Modifiers

1) Access Modifiers / Scope

2) Non-access modifiers

ACCESS MODIFIER:

In java we have four different Access modifier / scope:

1) private

2) default

3) protected

4) public

▶ private Modifier :

- private is an Access Modifier
- Any member which is prefixed with private modifier is known as private Modifier
- private member can be used only inside same class
- It can't be used in different class or sub class present in same package
- It can't be used in different class or sub class present in different package

▶ default Modifier / Package Scope:

If a member in java is not prefixed with any access modifier then it is known as default Scope.

A member with package scope can be used within same class, different class and subclass of same package

It cant be used either in different class, or sub class of different package

▶ protected Modifier:

Any member which is prefixed with protected Modifier is known as protected Modifier.

A protected member can be used anywhere within the package

It cant be used in different class of different package

It can be used outside the package but only with the help of sub-class.

NOTE: We can't make class protected

▶ public Modifier:

- Any member which is prefixed with public modifier is known as public Modifier.
- public member can be accessed anywhere within the package & outside the package

USAGE OF ~~KEYWORDS~~ MODIFIERS :

KEYWORDS	private	default	protected	public
Same class Same package	✓	✓	✓	✓
Different class Same package	✗	✓	✓	✓
Different class Different package	✗	✗	✗	✓
Sub class Different package	✗	✗	✓	✓

KEYWORDS	class	Variable	Methods	Constructor	Initialization ex.
private	✗	✓	✓	✓	✗
default	✓	✓	✓	✓	✗
protected	✗	✓	✓	✓	✗
public	✓	✓	✓	✓	✗